



The Linear Parametrization of Two Dimensional Tori Families in the Restricted Three Body Problem

Ian M. Down¹ · Kathleen C. Howell² · Manoranjan Majji¹ · Kyle T. Alfriend¹

Received: 2 April 2025 / Accepted: 16 December 2025

© The Author(s), under exclusive licence to American Astronautical Society 2026

Abstract

This paper analyzes the relationship between different family branches of 2-dimensional quasi-periodic invariant tori and their linear parametrization in both the circular and elliptical restricted three-body problems. Convergence basins associated with each family branch are defined by the maximal linear parametrization magnitude ε and computed via flow map reduction under rotational invariance. Convergence basins are correlated with tori stability indices and computational failure modes at the basin edge. Sensitivities of the convergence basin with respect to the dynamical system parameters of non-dimensional mass μ , and eccentricity e are examined. Invariant curve families expose geometric differences in tori family branches. 2-dimensional tori in the vicinity of common L_1 and L_2 libration point orbit families are analyzed. Regions where particular tori family branches are better represented by the linear parametrization are found, and present an important outcome of this study.

Keywords Invariant tori · Invariant curve · Quasi-periodic · Dynamical systems

✉ Ian M. Down
idown@tamu.edu

Kathleen C. Howell
howell@purdue.edu

Manoranjan Majji
mmajji@tamu.edu

Kyle T. Alfriend
alfriend@tamu.edu

¹ Department of Aerospace Engineering, Texas A&M University, 400 Bizzell St, College Station, TX 77840, USA

² School of Aeronautics and Astronautics, Purdue University, 610 Purdue Mall, West Lafayette, IN 47907, USA

1 Introduction

Quasi-periodic orbits (QPOs) are enticing trajectories in the context of multi-body spacecraft mission design because they are fundamentally bounded in the phase space, and exist in greater usable quantities compared to their periodic orbit counterparts in simplified astrodynamical models. This dramatically increases the preliminary design space. QPOs reside on the surfaces of quasi-periodic invariant tori (QPITs), and can be found in the vicinity of particular subfamilies of periodic orbits that radiate a complex center subspace. They were first investigated analytically by Farquhar and Kamel [1], with a numerical targeting algorithm later derived by Howell and Pernicka [2]. Gómez et al. [3, 4] as well as Jorba and Masdemont [5] then conducted rigorous explorations of these structures in the context of dynamical systems theory. In the process, several semi-analytical and computational algorithms related to the numerical determination of QPOs were developed. Despite continued investigation for a decade, QPOs and QPITs were largely an academic endeavor. It wasn't until robust algorithms were developed alongside an increase in general baseline computing power that these structures began consideration and integration into the preliminary mission design process. Because more QPOs exist at a specific energy level or particular frequency than periodic orbits, there is a greater possibility that one would satisfy the same mission constraints with lower cost (i.e. propellant or time). For example, heteroclinic connections are more prevalent between families of QPITs than families of periodic orbits, giving rise to cheaper access to more regions of cislunar space [6, 7]. Higher dimensional tori are also attractive structures on which to place multiple spacecraft in the context of formation flying and/or constellations. They can be phased in multiple directions corresponding to a torus' frequencies while still exhibiting bounded relative motion [8–13].

Tori of 2, 3, and 4-dimension are known to exist in simplified models of the Earth-Moon system. In the circular restricted three body problem (CR3BP), 2 and 3-dimensional tori exist in 2 and 3 parameter families, respectively. The elliptical restricted problem three body problem (ER3BP) has the potential for 4-dimensional tori, but sees a parameter family reduction of one because one torus frequency must coincide with the newly introduced periodic forcing frequency. Holding one (or more) of these parameters constant, family branches of QPITs can be grown from a specific periodic orbit with a complex center subspace. Many tools have been developed to examine and explore different combinations of these parameter branches [14], as well as to target a particular frequency set within a QPIT family [15]. When computing these structures, it is beneficial to reduce an invariant torus down to its *invariant curve*: the curve defined by a stroboscopic map that is invariant under the torus' turn angle(s). Despite the existence of many QPIT branches and hence invariant curve branches, there exists only one unique linear parametrization to these invariant curves that is used to initialize all QPITs in computation. The relationship between a particular family branch of tori and its linear parametrization in astrodynamical systems is so far unexplored.

This work examines the connections between branches of 2-dimensional invariant tori and their linear parametrization in the vicinity of libration point orbit families in both the CR3BP and ER3BP. The convergence basin of a particular family

branch given the initializing linear parametrization is used to quantify how effective the approximation is at representing the structure as it grows away from the periodic orbit. Specifically, constant energy, constant longitudinal frequency, constant latitudinal frequency, and constant frequency ratio family branches are explored for both L_1 and L_2 Lyapunov, halo, and vertical periodic orbit families in the Earth-Moon system. Polynomial fits of averaged basin measures expose the nonlinear disparities between families. Then, the 3:1 sidereal resonant orbits in both halo families are singled out, and the effect of varying the mass parameter μ on convergence basin is investigated. Lastly, all L_2 sidereal resonant halo orbits in the ER3BP corresponding to closure after one primary period are investigated in the context of varying the primaries' eccentricity e . Results illuminate regions along the orbit families where certain QPIT branches dominate as "most linear," and where others fail to exist at all. This paper is organized as follows. First, the restricted models of the three body problem are introduced. Next, QPITs and their computation are discussed in the context of astrodynamics. Then, the analysis carried out in this work is presented. Finally, results for the above orbits and their tori are shown and discussed.

2 The Restricted Three Body Problem

This work investigates tori structures found in both the CR3BP and ER3BP. Many of these structures persist in higher fidelity models with additional forcing frequencies as well as ephemeris, and can be transitioned using a variety of computational techniques [16–18]. Locating these structures in higher fidelity models is difficult without computation first in lower fidelity models, making lower fidelity analysis very useful. The non-dimensional CR3BP dynamics are given in Eq. 1

$$\ddot{x} - 2\dot{y} = \frac{\partial U^c}{\partial x} \quad (1a)$$

$$\ddot{y} + 2\dot{x} = \frac{\partial U^c}{\partial y} \quad (1b)$$

$$\ddot{z} = \frac{\partial U^c}{\partial z} \quad (1c)$$

$$U^c = \frac{1}{2} (x^2 + y^2) + \frac{1-\mu}{d_1} + \frac{\mu}{d_2}, \quad \mu = \frac{m_2}{m_1 + m_2} > 0 \quad (2)$$

where $d_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}$ is the distance between a spacecraft and the first (larger) primary, $d_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2}$ is the distance between a spacecraft and the second (smaller) primary, μ is the non-dimensional mass parameter, and $(\dot{\cdot}) = \frac{d(\cdot)}{d\tau}$. Starting from linearized solutions of planar periodic orbits about the collinear libration points in the CR3BP, a diverse set of periodic orbit families can be generated using continuation methods over numerical targeting schemes that exploit

various types of symmetry. Through Floquet theory, bifurcations of periodic orbits are identified through the lens of fixed points in maps [19]. The CR3BP orbit families used in the tori convergence basin analysis for this paper are shown in Fig. 1. The longitudinal and latitudinal frequencies from the linear approximation of the complex center subspace of these orbit families are shown in Fig. 2, where linearly stable regions are identified. For this work, longitudinal frequencies ω_0 correspond to the underlying periodic orbit, while the latitudinal frequencies ω_1 correspond to the additional frequency of a 2-dimensional QPIT in a periodic orbit's vicinity, and $(\tilde{\cdot})$ denotes quantities regarding the linear parametrization. Both the L_1 and L_2 halo families contain stable subsets where 3-dimensional QPITs exist. The periodic orbit center manifold that appears first in the family is denoted center manifold A (CM-A), while the second is denoted center manifold B (CM-B). This terminology is adopted from Lujan and Scheeres [20]. Note that only the L_1 Halo family has a non-monotonic frequency space for any given branch of frequencies presented here.

When including the eccentric motion of the primary bodies, periodic orbits in the restricted problem no longer exist in 1 parameter families, but only as isolated solutions with frequencies resonant with the periodic forcing frequency [21, 22]. The

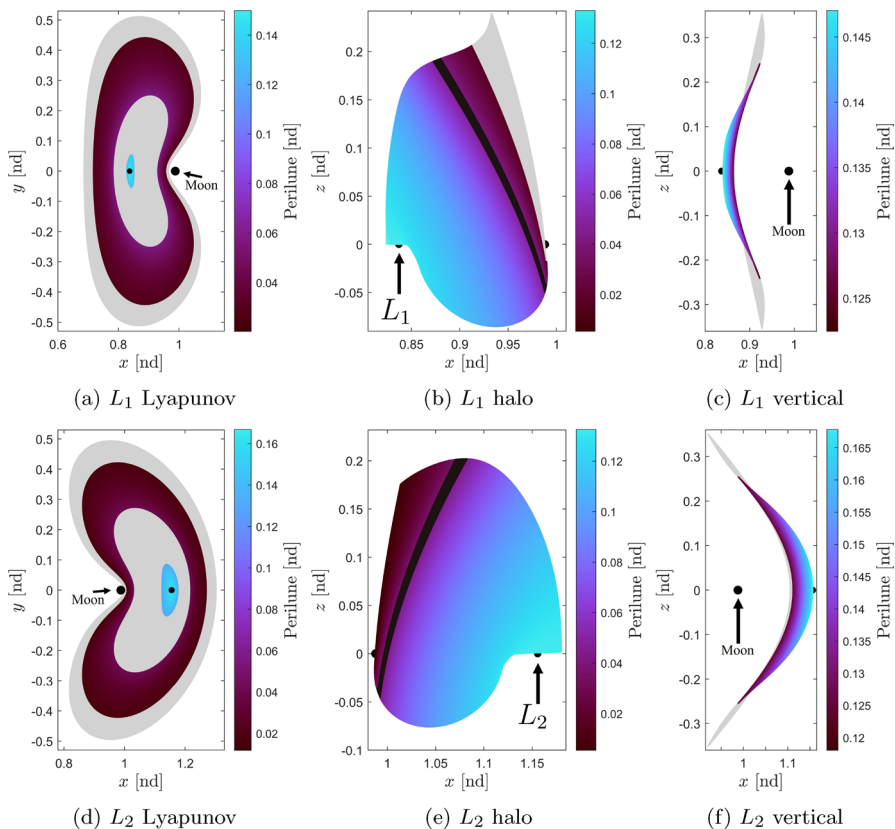


Fig. 1 Earth-Moon collinear libration point orbit families in the CR3BP

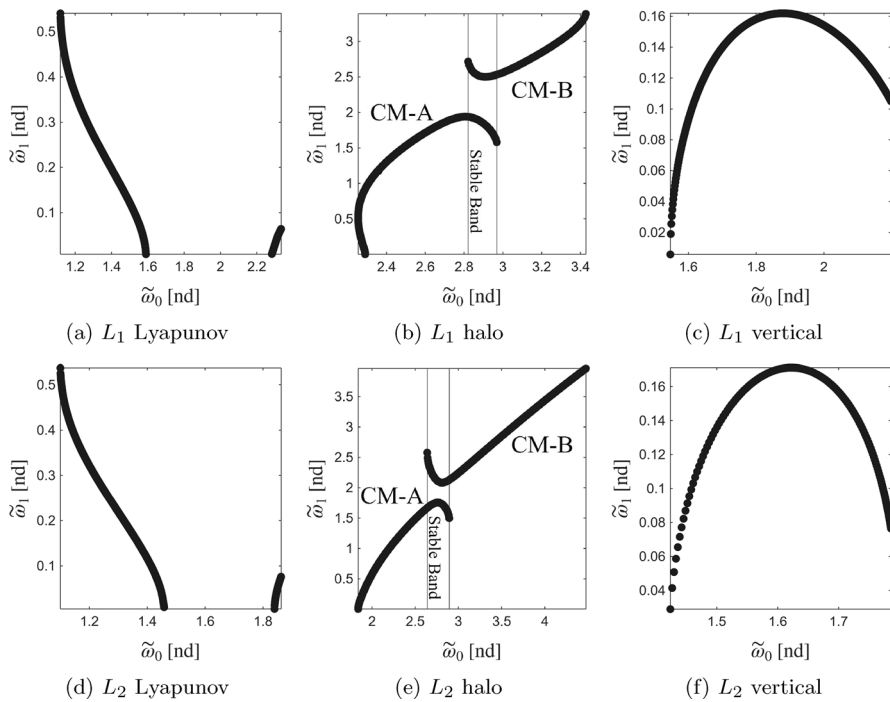


Fig. 2 Linear parametrization of the 2-dimensional tori frequency space for Earth-Moon collinear libration point orbit families in the CR3BP. All frequencies are in [rad/TU].

non-dimensional ER3BP dynamics, formulated with the primaries' true anomaly f as the independent variable are given in Eq. 3

$$x'' = 2y' + \frac{\partial U^e}{\partial x} \quad (3a)$$

$$y'' = -2x' + \frac{\partial U^e}{\partial y} \quad (3b)$$

$$z'' = \frac{\partial U^e}{\partial z} \quad (3c)$$

$$U^e = \frac{1}{1 + e \cos f} \left[\frac{1}{2}(x^2 + y^2 - ez^2 \cos f) + \frac{1 - \mu}{d_1} + \frac{\mu}{d_2} \right] \quad (4)$$

where $e > 0$ is the eccentricity of the primaries' motion and $(\cdot)' = \frac{d(\cdot)}{df}$. The period of the primaries' motion is then 2π , and $p : q$ *sidereal* resonant orbits are defined as those with p orbit revolutions per q sidereal months (i.e. number of times the Moon orbits the Earth). To compute sidereal resonant orbits in the ER3BP, solutions are initially computed in the CR3BP by locating orbits with periods $T = 2\pi \frac{q}{p}$, and then

continued over e as a parameter up to the desired value. The gravitational parameter and eccentricity of the Earth-Moon system are taken as 0.01215 and 0.0549, respectively, in this work.

Once periodic orbits are found numerically, their associated stable, unstable, and center subspaces can be explored through the use of the monodromy matrix M . The monodromy matrix is a linear approximation to the stroboscopic Poincaré map of trajectories neighboring a periodic orbit, which itself appears as a fixed point on the hypersurface [23]. The monodromy matrix is the state transition matrix (STM) integrated alongside a periodic orbit through one orbital period. The STM $\Phi = \frac{\partial \mathbf{x}(t)}{\partial \mathbf{x}(t_0)}$ satisfies the following relationships in Eqs. 5 and 6, with the monodromy matrix then as Eq. 7 where T is the orbital period.

$$\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (5)$$

$$\Phi(t_0, t_0) = \mathbb{I}_{6 \times 6} \quad (6)$$

$$M = \Phi(T + t_0, t_0) \quad (7)$$

The linear, time varying map $A(t)$ is obtained from the dynamics Jacobian of Eqs. 1 and 3 in phase space form, and is evaluated at every time t along the trajectory. It has the following structure.

$$A(t) = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ U_{xx} & U_{xy} & U_{xz} & 0 & 2 & 0 \\ U_{xy} & U_{yy} & U_{yz} & -2 & 0 & 0 \\ U_{xz} & U_{yz} & U_{zz} & 0 & 0 & 0 \end{bmatrix} \quad (8)$$

Linear approximations to a periodic orbit's associated local invariant manifolds are found by computing the eigenvalues and eigenvectors of M . These eigenvalues are also called characteristic, or Floquet multipliers. If a multiplier λ exists outside the unit circle in the complex plane, it is hyperbolic and belongs to either the stable or unstable subspace such that $|\lambda| < 1$ is stable, and $|\lambda| > 1$ is unstable. The global nonlinear hyperbolic manifolds are then tangent to their linear approximations at the fixed point of the stroboscopic map. Multipliers such that $|\lambda| = 1$ belong to the center subspace in the linear sense. Multipliers for Hamiltonian systems come in conjugate, reciprocal pairs [23]. Because the CR3BP is an autonomous (and conservative) Hamiltonian system and poses an energy integral, a trivial pair of unitary multipliers exists on the real line in the complex plane whose corresponding directions point along the orbit tangent and orbital family tangent. Complex unitary multipliers in Hamiltonian systems indicate the presence of quasi-periodic motion in the vicinity of periodic orbits. Their associated eigenvectors will be later exploited to construct a linear parametrization for the invariant curves of neighboring higher dimensional tori for a given point along the periodic orbit.

3 Quasi-periodic Invariant Tori

This section introduces quasi-periodic invariant tori (QPITs) in the context of astrodynamics. Their presence in families is considered for 2-dimensional tori in the CR3BP and ER3BP, and reduction to an invariant curve is shown. The linear parametrization of this invariant curve is presented, and computation of these structures is discussed. Finally, the stability computation of QPITs are given.

3.1 Existence in Astrodynamics

In the general sense, a p -dimensional torus has p independent frequencies of motion, and thus degrees of freedom. A torus is classified as a QPIT when two or more of these frequencies are incommensurate, or non-resonant, and the torus is invariant under the dynamical system's flow. A QPO is simply a trajectory bound to the surface of a QPIT. For an arbitrary set of initial angles $\theta(t_0)$ on a QPIT, it would take infinite time to return back to $\theta(t_0)$, and the resulting trajectory would densely cover the torus' surface in the process. As proved by Kolmogorov-Arnold-Moser (KAM) Theory applied to autonomous Hamiltonian systems, p -dimensional tori (if they exist), exist in $p + r$ parameter families, where $p \leq \bar{n}$, $\bar{n}$ is the number of generalized coordinates, and r is the number of free external parameters. Because of behavior near resonances between a torus' frequencies, families of tori in reality belong to Cantor sets that have exponentially small gaps near these resonances [24]. For a fixed value μ , the CR3BP can, and in fact does, boast 2 and 3-dimensional QPITs in 2 and 3 parameter families, respectively. When analyzing the ER3BP, a periodically forced, non-autonomous dynamical system, in the context of KAM Theory, it is appropriate to consider the periodic forcing phase as an additional state variable such that the minimum and maximum dimension of tori to exist is increased by one. For tori to exist in this system, at least one frequency must be commensurate with the forcing term, hence family dimension is reduced by one [25]. Thus for fixed values μ and e , the ER3BP in contrast boasts 2, 3, and 4-dimensional QPITs in 1, 2, and 3 parameter families, respectively.

In examining the center subspace of periodic orbits (1-dimensional tori), 2-dimensional QPITs are found to be more prevalent as is evident in Fig. 1, but 3-dimensional tori do exist, for example, around some halo and distant retrograde orbits in the CR3BP. This work focuses solely on 2-dimensional tori because of their presence around a wider variety of periodic orbits in lower fidelity models of the three body problem. For analysis and insight into 3-dimensional QPITs in the CR3BP see [20, 26].

3.2 2-Dimensional Tori

Tori of dimension 2 are now examined in detail. Higher dimensional tori structures are analogous, albeit harder to visualize and less common in the circular and elliptical restricted three body problems. Let a diffeomorphism be defined as a continuously differentiable, bijective function with a continuously differentiable inverse. A torus function Γ is then the diffeomorphism that maps angles on the standard p -torus to a

dynamical system's n -dimensional phase space, hence embedding a particular torus in that dynamical system.

$$\Gamma : \mathbb{T}^p \rightarrow \mathbb{R}^n \quad (9)$$

For this torus to be invariant under the dynamical system's flow, i.e. vector field, the following set of invariance conditions must be met

$$\frac{d}{dt}\{\Gamma = \mathbf{x}\} = \sum_{i=1}^p \dot{\theta}_i \frac{\partial \Gamma}{\partial \theta_i} = \mathbf{f}(\Gamma, t), \quad \theta_i \in [0, 2\pi) \quad (10)$$

where $\mathbf{f}$ is the vector field of, in general, a quasi-periodically forced dynamical system. If external forcing frequencies are present in a dynamical system like the ER3BP, the set or a subset of torus frequencies must be resonant with all external frequencies to exist [24]. Internal torus frequencies are then any additional torus frequencies that do not correspond to external forcing terms. Any θ_i that corresponds to an internal frequency has a rate of change $\dot{\theta}_i$ that is not uniquely defined. In this case, it is most common to choose $\dot{\theta}_i = \omega_i$, a constant, so the relationship

$$\theta_i(t) = \omega_i(t - t_0) + \theta_i(t_0) \quad (11)$$

holds, where ω_i are the (sub)set of fundamental torus frequencies. Thus, Γ is uniquely defined as the fundamental torus function given a particular set of initial conditions $\theta(t_0)$ in angle space with respect to those on the torus in the phase space $\mathbf{x}(t_0)$.

A 2-dimensional torus is governed by two independent frequencies, ω_0 and ω_1 . These frequencies are often termed the longitudinal frequency and latitudinal frequency, respectively, because ω_0 typically corresponds to an underlying periodic orbit. Related to these frequencies are the longitudinal period, or mapping time T_0 and the rotation number, or turn angle ρ .

$$T_0 = \frac{2\pi}{\omega_0} \quad (12)$$

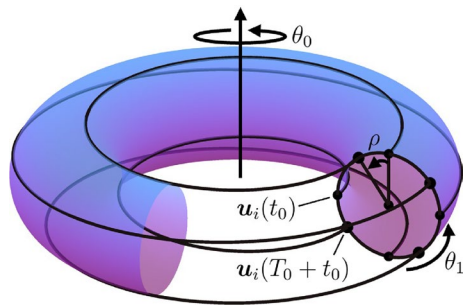
$$\rho = T_0 \omega_1 \quad (13)$$

The turn angle ρ is the latitudinal angle a particle on the torus twists after T_0 . This is shown graphically in a topological representation of a 2-dimensional torus in Fig. 3.

Any p -dimensional QPIT can be reduced to a $(p - 1)$ -dimensional torus through the lens of a stroboscopic map. For $p = 2$, the resulting reduction is known as an *invariant curve*, shown as the black circle in Fig. 3. The invariant curve is defined by a torus' intersection with a stroboscopic map with period T_0 such that states $\mathbf{u}_i$ on the curve obey the dimensionally reduced rotational flow map invariance condition

$$\mathbf{R}(-\rho)\mathbf{u}_i(T_0 + t_0) - \mathbf{u}_i(t_0) = \mathbf{0} \quad (14)$$

Fig. 3 Topological representation of a 2-dimensional torus with a discretized invariant curve and a quasi-periodic orbit



where $\mathbf{R}(-\rho)$ is the rotation operator. Given the invariance under the flow, a torus is fully represented at any time t by its invariant curve.

The linear parametrization, and hence approximation, to a torus' invariant curve is derived from the discrete linear solution of the complex center subspace of a period orbits' monodromy matrix $\mathbf{M}$. The periodic orbit itself appears as a fixed point in this linearized stroboscopic map. The subspace corresponding to a 2-dimensional QPIT has complex conjugate eigenvalues and eigenvectors $\lambda_{C,1} = \lambda_{C,2}^*$ and $\mathbf{v}_{C,1} = \mathbf{v}_{C,2}^*$, where $|\lambda_C| = 1$. Because of conjugacy, only one eigenpair needs to be considered. By only entertaining initial conditions in the complex center subspace, the linear solution of this discrete map takes the following phase-shifted form [4, 27],

$$\begin{aligned} \tilde{\mathbf{u}}^{(m)} &= \varepsilon |\lambda_C|^m (\operatorname{Re}[\mathbf{v}_C] \cos(m\tilde{\rho} + \theta) - \operatorname{Im}[\mathbf{v}_C] \sin(m\tilde{\rho} + \theta)) \\ \tilde{\rho} &= \tan^{-1} \left(\frac{\operatorname{Im}[\lambda_C]}{\operatorname{Re}[\lambda_C]} \right) \end{aligned} \quad (15)$$

where ε and θ are the magnitude and phase of an initial condition vector, respectively, and m is the iteration count over the map. The quantity $\tilde{\rho}$ is the turn angle for the linear discrete map, and hence acts as a linear approximation to the turn angle under the nonlinear flow. Note that the choice of either $\lambda_{C,1}$ or $\lambda_{C,2}$ changes the value $\tilde{\rho}$ when quadrants in the complex plane are considered. This then directly affects the resulting latitudinal frequency defined in Eq. 13. Either choice is a valid parametrization of the subspace and simply offers a different perspective of the same structure because of opposing directions of the imaginary components of the complex conjugate eigenvectors. In other words, there is inherent ambiguity in the initial coordinate system used to define the invariant curve. Tori grown out of these parametrizations for the same underlying periodic orbit are equivalent structures, as shown for the Earth-Moon L_2 halo family in [28]. Setting $m = 0$ and holding a constant magnitude ε , a discretized representation of an entire invariant curve relative to a period orbits' fixed point $\mathbf{x}_F$ is achieved by considering a uniform distribution of N initial phases. This is expressed in Eq. 17. Refer to Fig. 3 for a visual representation.

$$\theta_{1i} = \frac{2\pi(i-1)}{N}, \quad i \in [1, N] \quad (16)$$

$$\tilde{\mathbf{u}}_i = \varepsilon (\operatorname{Re}[\mathbf{v}_C] \cos \theta_{1i} - \operatorname{Im}[\mathbf{v}_C] \sin \theta_{1i}) \quad (17)$$

$$\tilde{\mathbf{x}}_i = \tilde{\mathbf{u}}_i + \mathbf{x}_F \quad (18)$$

Note that the discretized linear approximation is the equation of an ellipse in the phase space with the periodic orbits' fixed point at the center. Lastly, the approximation of the stroboscopic mapping time is simply the period of the underlying orbit, such that $\tilde{T}_0 = T$. Note that an approximation to an entire 2-dimensional torus results from the collection of invariant curve approximations spanning the periodic orbit.

Family Branches of Invariant Tori. In the CR3BP, 2-dimensional tori lie in two parameter families. However, to understand the structure of these families, it is more convenient to define a scalar constraint as a function of these two parameters, and explore one parameter subsets of the family. Each subset is termed a *branch*. There are an infinite number of choices for such a function, but four that arise naturally and are hence examined in this work are:

$$\begin{aligned} \text{Constant Energy: } & JC = \tilde{J}C \\ \text{Constant Longitudinal Frequency: } & \omega_0 \propto 1/T_0 = 1/\tilde{T}_0 \\ \text{Constant Latitudinal Frequency: } & \omega_1 = \rho/T_0 = \tilde{\rho}/\tilde{T}_0 \\ \text{Constant Frequency Ratio: } & \omega_1/\omega_0 \propto \rho = \tilde{\rho} \end{aligned} \quad (19)$$

where JC is Jacobi's Constant, and the $(\tilde{\cdot})$ values are those computed from the periodic orbit from which the 2-dimensional tori emanate. Notably, the constant ρ branch was examined in the foundational work by Gómez and Mondelo in their investigation into the center manifold of collinear libration points in the restricted problem [4]. Despite the fact that many family branches can be constructed, only one first order linear parametrization, and hence approximation, in Eq. 17 exists to approximate them all. Geometrically examining and comparing these one parameter branches as tori projected into configuration space is difficult because each torus appears as a 2-dimensional surface. However, when reduced to their invariant curves, geometric differences become tractable and are more easily observed.

3.3 Computation

Many methods have been derived to numerically compute quasi-periodic invariant tori in dynamical systems, a handful of which are particularly suited for astrodynamics. Methods that leverage normal forms can be used to locate explicit motion directly on the invariant surfaces of elliptic tori [29], though this requires heavy use of algebraic manipulation software. Some methods aim to compute a solution to the set of flow invariance partial differential equations defined in Eq. 10 directly, such as Schilder et al. [30] via finite central differencing, or discrete Fourier transforms [31]. Refining the mesh used in computation to achieve desired accuracy with the entire torus surface represented can prove numerically unmanageable. For that reason,

computational methods have been developed that track only a torus' invariant curve or higher dimensional equivalent. One such method uses spatially defined Poincaré maps to solve for the invariant curve as a two point boundary value problem (TPBVP) [32]. Another, developed originally by Gómez and Mondelo [4] and improved on by Olikara and Scheeres [33], then McCarthy [26], enforces rotational invariance under the stroboscopic flow map. This algorithm is typically referred to as GMOS in literature, after its contributors. The algorithm was recently further improved upon by Fernández-Mora, Haro, and Mondelo to more efficiently compute 3 and higher dimensional tori in quasi-periodically forced Hamiltonian systems [34]. Separate improvement is made to the computational complexity of the stroboscopic map approach through the use of quasi-Newton methods [35]. In general, the stroboscopic map approach trades a reduction in parameter space for a less global representation of the structure during computation. In cases of high nonlinearity or long stroboscopic mapping times, this may diminish solution convergence. Baresi et al. [31] conducted a study that compared the accuracy of several of the aforementioned computational methods, and concluded that GMOS offers superior accuracy at a lower computational cost. The original algorithm later detailed below is also straightforward, and quick to implement and initialize, for example with the variational space of periodic orbits. This accessibility has lead to widespread use in research. As such, the original adaptation of GMOS is used as the method of computation in this analysis.

The GMOS algorithm comes in two flavors: shooting and collocation. The multiple shooting method is used in this work rather than single shooting so that results obtained have less dependence on where a singular invariant curve is initialized around a given periodic orbit. The single shooting method is briefly introduced below for compactness, but an in depth derivation of multiple shooting in matrix form is nicely formulated in [26]. To begin, the TPBVP's free variables consist of N states on a discretized invariant curve relative to a periodic orbit's fixed point, along with the torus' mapping time and turn angle.

$$\mathbf{X} = \begin{bmatrix} \mathbf{u}_1(t_0) \\ \vdots \\ \mathbf{u}_N(t_0) \\ T_0 \\ \rho \end{bmatrix}, \quad \mathbf{u}_i = \begin{bmatrix} u_{x,i} \\ u_{y,i} \\ u_{z,i} \\ u_{\dot{x},i} \\ u_{\dot{y},i} \\ u_{\dot{z},i} \end{bmatrix} \quad (20)$$

The problem's constraints consist of the enforcement of the rotational invariance condition, along with a family branch constraint F_B corresponding to Eq. 19.

$$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} \mathbf{R}(-\rho)\mathbf{u}_1(T_0 + t_0) - \mathbf{u}_1(t_0) \\ \vdots \\ \mathbf{R}(-\rho)\mathbf{u}_N(T_0 + t_0) - \mathbf{u}_N(t_0) \\ F_B \end{bmatrix} = \mathbf{0} \quad (21)$$

If computing a constant energy torus, the average value of Jacobi's Constant over the invariant curve is used in the constraint.

$$\overline{JC} = \frac{1}{N} \sum_{i=1}^N JC_i \quad (22)$$

The rotation matrix $\mathbf{R}$ required to compute the invariance constraint is constructed using the discrete Fourier transform. The linear approximation above is used to initialize a solution. With $\mathbf{X} \in \mathbb{R}^{6N+2}$ and $\mathbf{F} \in \mathbb{R}^{6N+1}$, the minimum norm update of a first order Newton's method is used to iterate towards convergence for this under-constrained system.

$$\mathbf{X}_{j+1} = \mathbf{X}_j - \mathbf{D}\mathbf{F}^T(\mathbf{X}_j) [\mathbf{D}\mathbf{F}(\mathbf{X}_j) \mathbf{D}\mathbf{F}^T(\mathbf{X}_j)]^{-1} \mathbf{F}(\mathbf{X}_j) \quad (23)$$

The constraint Jacobian $\mathbf{D}\mathbf{F}$ is dense, has a 1-dimensional null space, and its pseudo-inverse is well computed using QR decomposition. Exploiting the null space dimension, pseudo-arc length continuation (PALC) is employed to continue tori along a family branch using the tangent space of the branch's manifold. Let $\mathbf{X}_{j-1}^*$ be a previous solution, $\Delta\mathbf{X}_{j-1}^*$ be the previous solution's constraint Jacobian null space, and δs be an arc length step in the solution space. The initial guess for the next branch's free variable vector is

$$\mathbf{X}_j = \mathbf{X}_{j-1}^* + \delta s \Delta\mathbf{X}_{j-1}^* \quad (24)$$

The constraint vector is then augmented to ensure any new converged solution is one arc length step away from the previous solution.

$$\mathbf{G}(\mathbf{X}_j) = \begin{bmatrix} \mathbf{F}(\mathbf{X}_j) \\ (\mathbf{X}_j - \mathbf{X}_{j-1}^*)^T \Delta\mathbf{X}_{j-1}^* - \delta s \end{bmatrix} = \mathbf{0} \quad (25)$$

$$\mathbf{D}\mathbf{G}(\mathbf{X}_j) = \frac{\partial \mathbf{G}(\mathbf{X}_j)}{\partial \mathbf{X}_j} = \begin{bmatrix} \mathbf{D}\mathbf{F}(\mathbf{X}_j) \\ [\Delta\mathbf{X}_{j-1}^*]^T \end{bmatrix} \quad (26)$$

When continuing down branches of tori using GMOS, there is the possibility for an invariant curve solution to be simply a different invariant curve on the previously computed torus, leading to different parametrizations of the same structure. To encourage convergence to a new, unique solution, two additional constraints are augmented to the system that take away this phase ambiguity by extremalizing the $\mathcal{L}_2$ norm between the current invariant curve and the previous solution. When coupled with the PALC constraint, this forces the extremal point to be a maximum under a particular arc length step. These phase constraints take the form of the following vector inner products.

$$F_{\theta_0} = \sum_{i=1}^N \langle \mathbf{u}_i, \frac{\partial \mathbf{u}_i^*}{\partial \theta_0} \rangle \quad (27)$$

$$F_{\theta_1} = \sum_{i=1}^N \langle \mathbf{u}_i, \frac{\partial \mathbf{u}_i^*}{\partial \theta_1} \rangle \quad (28)$$

Sensitivities of the discretized invariant curve points with respect to the torus angles are computed via the discrete Fourier transform and the vector field invariance condition [26].

Adding two phase constraints on top of the PALC constraint results in a now over-determined TPBVP, a typical occurrence for many Hamiltonian systems [33]. Still, the null space of the constraint Jacobian remains 1-dimensional, and branch continuation can proceed, though the least-squares update is now used in place of the minimum norm update in Eq. 24. If one is concerned with isochronous comparisons between QPO trajectories on different QPITs such that initial conditions on the invariant curve begin at exactly the same phase for a particular fixed point, symmetry constraints can be used in place of phase constraints for invariant curves who have inherent symmetry with respect to the phase space. For example, invariant curves with a fixed point on the xz -plane of a symmetric, collinear libration point orbit will also exhibit this symmetry. Using symmetry constraints in place of phase constraints requires smaller arc length steps, and is generally less robust, but may be suited for particular analysis such as spacecraft formation design [36].

3.4 Stability

One particular advantage of using a shooting algorithm over collocation in the computation of QPITs is the ease at which stability information can be extracted for a particular solution. Using the sub-block of the shooting Jacobian corresponding to the invariance condition, the following stability matrix $\mathbf{P}$ is defined

$$\frac{\partial(\mathbf{R}(-\rho)\mathbf{u}(T_0 + t_0) - \mathbf{u}(t_0))}{\partial \mathbf{u}} = \mathbf{P} - \mathbb{I}_{6N \times 6N} \quad (29)$$

$$\mathbf{P} = (\mathbf{R}(-\rho) \otimes \mathbb{I}_{6 \times 6}) \mathbf{\Psi} \quad (30)$$

with $\otimes$ denoting the Kronecker product. The matrix $\mathbf{\Psi}$ is a sparse matrix consisting of each invariant curve point's STM Φ propagated up to T_0 .

$$\mathbf{\Psi} = \begin{bmatrix} \Phi_1(T_0 + t_0, t_0) & \mathbf{0}_{6 \times 6} & \cdots & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \Phi_2(T_0 + t_0, t_0) & \cdots & \mathbf{0}_{6 \times 6} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \cdots & \Phi_N(T_0 + t_0, t_0) \end{bmatrix} \quad (31)$$

The eigenstructure of $\mathbf{P}$ is then used to characterize the stability of a particular QPIT in the same fashion as $\mathbf{M}$ is used to compute the stability of a periodic orbit. The number of Floquet multipliers obtained from $\mathbf{P}$ in the case of a 2-dimensional torus is $6N$, and each exists on one of three concentric circles in the complex plane [26]. This is similar to the fact that $\mathbf{M}$ has only three unique Floquet multipliers characterizing

stability due to the symplectic nature of the CR3BP and ER3BP. A stability index is then defined as the maximum multiplier of P .

$$\nu = \max(|\lambda_i|) \quad (32)$$

4 Analysis

The goal of this work is to analyze the connection between invariant tori and their linear approximation for common tori family branches in the restricted three body problem. In other words, how nonlinear is a converged invariant torus compared to its approximation? One choice to measure this relationship is to use finite time Lyapunov exponents (FTLEs) to form a nonlinear index [37]. Here, the normalized state transition matrix difference between points on the invariant curve solution and its linear approximation are integrated to the torus solution's mapping time, and the maximal, relative FTLE is computed. This isochronous metric was used in previous work by the author [36], and showed variety in the best approximated torus family based on the periodic orbit family in question. A similar metric could be derived with higher order FTLEs computed by integrating higher order state transition tensors that capture more nonlinearity [38]. These methods rely on empirical sampling; there is ambiguity in what value ε , the linear approximation offset in Eq. 17, to choose in coordination with a converged solution. Additionally, converged invariant curves of QPITs in the CR3BP can wander and rotate slightly with respect to their fixed point, making direct comparison between a point on the invariant curve and its approximation difficult and somewhat arbitrary without forming an optimization problem. Symmetry constraints can be introduced in place of phase constraints as mentioned above, but the choice of ε still directly affects the relative FTLE results. Values could be averaged over the invariant curves, but performing this computation for multiple family branches of tori around multiple families of periodic orbits is intractable.

Since it is less desirable to compare individual points on invariant curves, and instead look at their approximating capability as a whole, a more appropriate metric is the associated *convergence basin*. That is, solve for the maximum value ε such that $\delta\varepsilon > \varepsilon$ results in algorithmic failure for a given torus family around a particular set of periodic orbit fixed points. Using multiple shooting via the computational method described above then gives a linear approximation for the entire structure, as opposed to a single invariant curve. In trajectory analysis, convergence basins for perturbations of free variables are shown to be an effective design tool [39]. Because the linear approximation of the invariant curves is parametrized using only one free parameter ε , which is sign invariant due to the rotational symmetry of the curve, only $\mathbb{R}^+$ of ε need be examined. Note that the convergence basin of a particular TPBVP is directly related to the solution method applied. Higher-order Newton solvers would typically yield a larger convergence basin than a traditional first order method because more nonlinearity is captured in each update. This can result in fewer steps to convergence, and depending on the problem's nonlinearity, actually less computation time [40]. A drawback is an increase in algorithmic complexity with the need for obtaining

higher-order information through analytical, semi-analytical, or numerical means, which can be computationally costly. This paper only considers the basin associated with the common, first order Newton's method using a multiple shooting scheme.

For a given family subset of a periodic orbits in Fig. 1 that exhibit quasi-periodic behavior in their vicinity, a convergence basin value ε is computed for each of the four torus family branches in Sect. 3.2, per periodic orbit in the CR3BP. Similar analysis follows in the CR3BP for a varied parameter μ . Lastly, the same analysis is conducted in the ER3BP for the varied parameter e . Solution convergence is determined when $\|\mathbf{F}\| < 1 \times 10^{-12}$. Equally as important, algorithmic failure is decomposed into three failure modes. First, *solution divergence* in the context of this computation is concluded when $\|\mathbf{F}\| > 1 \times 10^{-1}$. Constraint magnitudes above this are seen as unrecoverable by the algorithm. Second, *solution time out* as a failure mode was concluded when the Newton's method iteration count reached $j > 100$. Typically, solutions initialized close to the underlying structure and far from the convergence basin edge converge in under 10 iterations. Because of the initial solution's phase ambiguity, solutions are free to grow, shrink, wander, and rotate from the linear approximation they were initialized at during each iteration towards convergence. This behavior is usually very small when the linear approximation well approximates the torus. However, at the convergence basin edge defined for a particular solution method where the linear approximation is at its limits, this behavior is found to be exaggerated, with each iteration jumping between significantly different torus parametrizations and family members without triggering any other algorithmic failure modes. The value $j = 100$ was found to be an appropriate iteration cut-off because solutions that converged after this value typically required a far greater number of iterations than was acceptable in this study, for example $j > 200$. While this failure mode is seen across all tori family branches, time out is especially prevalent for the constant ω_1 family branches because the family constraint in the GMOS formulation relies on changing two sensitive parameters T_0 and ρ , rather than one. Lastly, the third failure mode is *solution collapse*, where a higher dimensional invariant torus collapses either onto its own underlying periodic orbit, onto a neighbor, or onto a singular point in the phase space with near zero stroboscopic mapping time T_0 and/or erroneous turn angle ρ that deviates far from the linear approximation. To numerically evaluate these criteria, only one invariant curve need be examined. To check collapse onto a periodic orbit, Eq. 33 is used,

$$\frac{1}{N} \sum_{i=1}^N \|\mathbf{u}_i\| < 1 \times 10^{-7} \quad (33)$$

to check collapse onto a neighboring periodic orbit, Eq. 34 is used,

$$\frac{1}{N} \sum_{i=1}^N \|\mathbf{u}_{i+1} - \mathbf{u}_i\| < 1 \times 10^{-7} \quad (34)$$

and to check mapping time or turn angle collapse, Eqs. 35 and 36 are used.

$$\frac{|T_0 - \tilde{T}_0|}{\tilde{T}_0} < 0.1 \quad (35)$$

$$\frac{|\rho - \tilde{\rho}|}{\tilde{\rho}} < 0.1 \quad (36)$$

The convergence basin value ε is refined using a method of bisections as in [41]. If $\varepsilon < 1 \times 10^{-7}$, it is concluded that the particular family branch of tori does not exist around that specific periodic orbit given the computational constraints and method imposed on this analysis. Hence, this tolerance is used to check solution collapse. This does not directly imply the torus does not exist within the dynamics themselves, as it is possible a different computational method or simply an increase in discrete torus points could be used for a successful initialization. However, as will be shown in the subsequent results, nonexistence predicted by the tolerance assumed in this paper aligns well with predicted nonexistence for tori initiated on or near internal system resonances (recall that families of QPITs exist as Cantor sets). Moreover, results presented below agree with the sparsity of constant ω_1 family branches found by Lujan and Scheeres [14].

Convergence basins are directly affected by the hyperbolic nature of the variational space [39]. To better understand this correlation, the resulting data is compared to the stability index ν of the respective torus as a function of family member. The GMOS multiple shooting algorithm is implemented using $M = 5$ discretized invariant curves comprised of $N = 30$ points each for all computations herein. Note that a majority of the QPITs computed in this paper can converge using fewer points and fewer curves. It is possible that using $N > 30$ invariant curve points may yield larger convergence basin measures because more complex geometries can be accurately represented with additional Fourier modes. However, this is beyond the scope of the analysis herein. The conclusions are drawn by comparing results relative to the computational constraints imposed. Lastly, is it worth noting that where along a particular periodic orbit an invariant curve's fixed point is placed in the context of GMOS also may affect the algorithm's convergence basin. For example, choosing to initialize GMOS at an NRHO's periapse could result in a smaller convergence basin when compared to initializing at the apoapse because the sensitivities are more pronounced where the rotational invariance condition is trying to be satisfied. This is alleviated with multiple shooting, though not eliminated. Fixed points used in the evaluation of the invariance condition for all halo and Lyapunov families reside at their respective apoapse perpendicular crossings with respect to the second, smaller primary where the dynamics are less sensitive. Fixed points used in the evaluation of the invariance condition for both vertical families reside at the orbits' x -axis crossing. At these fixed points, the invariant curves of the 2-dimensional tori exhibit the same symmetries as their underlying periodic orbit counterparts in configuration space.

Subsets of the L_1 and L_2 halo orbit families are elliptic, and admit 3-dimensional tori in their vicinity in the CR3BP, with an additional frequency ω_2 now present. The above analysis is carried out for the 2-dimensional tori $\{\omega_0, \omega_1\}$ (CM-A) and $\{\omega_0, \omega_2\}$ (CM-B) that are embedded in their higher dimensional structures when

these subsets are encountered. Similar notation is used for higher dimensional tori found in the ER3BP.

5 Results

This section presents tori convergence basin results via the parameter ε for orbits in Fig. 1 from the Earth-Moon CR3BP. The non-dimensional mass parameter μ is then varied for two sidereal resonant orbits in the halo families within the CR3BP to analyze the effects of the parameter on tori convergence basins. In particular, the L_1 3:1 halo and L_2 3:1 halo orbits are analyzed with a parameter sweep of $\mu \in [0.001, 0.05]$. These orbits were chosen because of their frequency match and existence around separate libration points. In addition, halo families have been shown desirable for mission design in various three body systems [42, 43]. The L_2 3:1 halo, along with 2:1 and 4:1 sidereal resonant halo orbits are then continued into the ER3BP and the eccentricity parameter $e \in [0.001, 0.1]$ is varied for a similar convergence basin analysis. The L_1 3:1 halo orbit is not included in the ER3BP analysis because tori in its vicinity are completely destroyed before reaching the Earth-Moon eccentricity of 0.0549. Previous authors have identified a sub-region of the L_2 halo family in the ER3BP that contains numerous fold bifurcations, making the transition of structures in this region to full ephemeris models difficult [17, 44, 45]. The L_2 3:1 halo orbit inhabits this region, with the 2:1 and 4:1 orbits existing outside the region from either side of the frequency spectrum. Hence, these three isolated orbits within the ER3BP compactly characterize the L_2 halo space. The L_2 3:1 halo orbit is distinctly interesting because it is fully elliptic in the ER3BP for a subset of e , meaning 4-dimensional QPITs exist here. It is important to reiterate that the convergence basin measures derived numerically in the below results are partially a function of the computational method and failure mode conditions defined in this analysis. Because these conditions are held constant throughout the analysis for all cases, insightful conclusions from the results are obtained.

5.1 Earth-Moon CR3BP Libration Point Tori

The convergence basin results for the Earth-Moon L_1 Lyapunov family are given in Fig. 4. All tori family branches emanating from the smaller subset of periodic orbits nearest L_1 have larger convergence basins than the larger subset with smaller perilune passes because these structures are further away from the singularity at the second primary (Moon). Constant ω_1 and constant ρ family branches have a significant and consistently larger convergence basin between perilune radii of 0.0517 and 0.055 [nd], with an average difference $\Delta\varepsilon = 0.021$ [nd]. This implies that the linear approximation better represents these family branches in this particular region. Constant JC and constant ω_0 family branches have larger convergence basins in tori generated from periodic orbits closest to L_1 . All tori family branches examined show a sharp decrease in convergence basin near $r_p = 0.0356$ [nd] corresponding to an underlying orbital frequency $\omega_0 = 1.2527$ and turn angle $\rho = 1.5647$. This is nearly the frequency resonance $\omega_1/\omega_0 = 1/4$, but curiously, collapse was not a failure mode

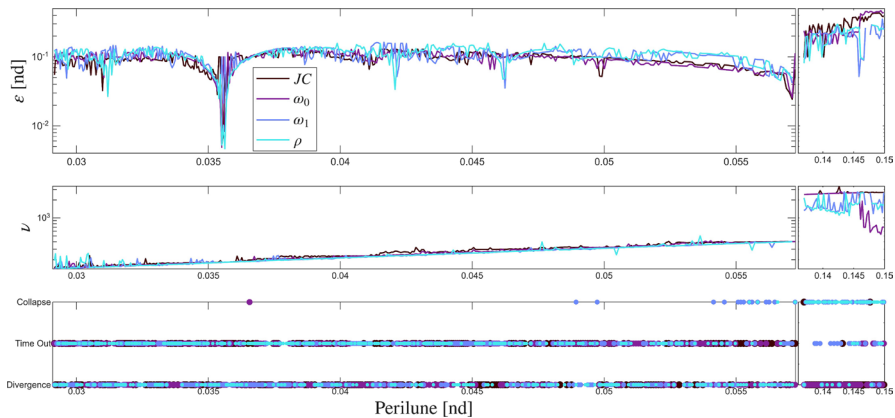


Fig. 4 Earth-Moon CR3BP L_1 Lyapunov invariant tori convergence basins

for any of these tori. The stability index for each QPIT largely matches the stability index of the underlying periodic orbit for the larger subset of L_1 Lyapunov orbits. However, this trend is abandoned for the subset closer to L_1 . Note that constant ω_1 family branches exist around nearly all periodic orbits analyzed in the L_1 Lyapunov family. This is not the case for the halo orbit families, as will be shown. Failure modes for the larger subset with smaller perilunes are more or less evenly distributed across time out and divergence. Collapse is the dominant failure mode for only the constant ρ family branch around the subset closer to L_1 . Tori family branches were continued out of a single L_1 Lyapunov orbit with frequency $\omega_0 = 1.5448$ and perilune $r_p = 0.0540$ [nd] to show the geometric differences in configuration space for a region where one family branch is significantly better represented by the linear approximation. Similar analysis is done for the other five orbit families. Tori are continued until the null space of the shooting Jacobian ceases to exist numerically, or a turning point in either T_0 or ρ is encountered. These invariant curve families around their respective fixed points detailed in Sec. 4 are shown in Fig. 11.

The convergence basin results for the Earth-Moon L_1 halo family are given in Fig. 5. Tori branches grown from L_1 halo orbits show more variety in their respective convergence basins. From the bifurcation with the Lyapunov family up until a perilune of 0.1 in CM-A, constant JC family branches show the largest convergence basins while constant ω_1 branches show the least. No constant ω_1 tori exist between perilunes of 0.0579 and 0.0850 within CM-A. CM-B sees the constant ρ branch best approximated by the linear subspace of the periodic orbits throughout. CM-B also sees no existence of constant ω_1 tori. This is qualitatively consistent with [28], whose results span the L_2 halo family. Both CM-A and CM-B decline in basin measure when they approach the stable band, which is shown with a zoomed view in Fig. 5. Within the stable band, CM-B tori display a larger convergence basin for all branches except constant ω_1 . The stability indices largely mirror the underlying periodic orbits'. Hyperbolic 2-dimensional tori are more readily present in the stable band of halos for tori on the edge of their convergence basin. This does not, however, capture the stability of each fully grown branch. CM-A sees the constant JC branch as slightly more unstable than all other branches, despite differing results for basin

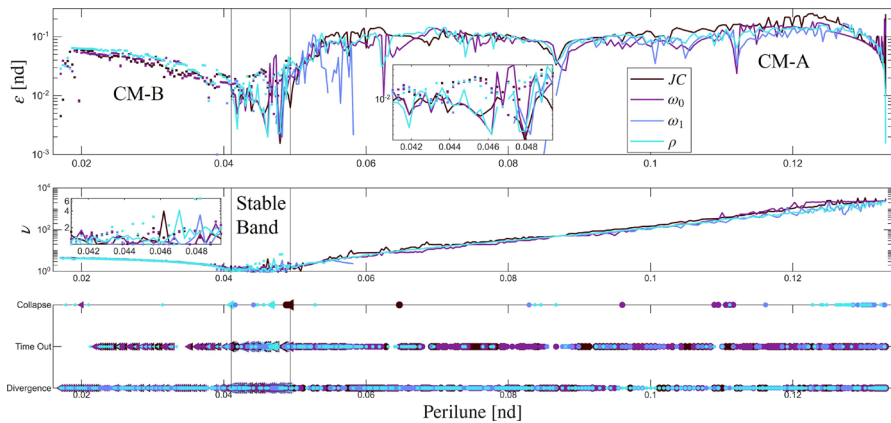


Fig. 5 Earth-Moon CR3BP L_1 halo invariant tori convergence basins

measure. Divergence is the dominant failure mode for constant ρ family branches, and ω_1 family branches when they don't exist. Time out is most commonly seen in constant JC branches, and when constant ω_1 branches do exist. This is largely because of the numerical method used: the independent variables T_0 and ρ used in the stroboscopic invariant curve approach must both change using little information from the jacobian for constant $\omega_1 = \frac{\rho}{T_0}$. This causes incremental and oscillatory Newton updates, and thus convergence takes a greater number of iterations when compared to the other family branches. Tori family branches were continued out of a single L_1 halo orbit with frequency $\omega_0 = 2.2599$ and perilune $r_p = 0.1073$ [nd], and are shown in Fig. 11.

The convergence basin results for the Earth-Moon L_1 vertical family are given in Fig. 6. The family exhibits several regions where constant ω_1 branches do not exist. These overwhelming correspond to the collapse failure mode. Only near L_1 between perilunes 0.1442 and 0.1465 is there a distinct convergence basin difference, where constant ρ and constant ω_1 have an averaged $\varepsilon = 0.2015$ compared to $\varepsilon = 0.0979$ of constant JC and constant ω_0 branches. Constant JC and constant ω_0 tori branches share the most similarity in stability index with the underlying vertical periodic orbit family. The constant ρ branches are largely more stable at the convergence basin edge and are dominated by the divergence failure mode. Tori of all family branches failed by collapse into the libration point itself at closest perilune. Tori family branches were continued out of a single L_1 vertical orbit with frequency $\omega_0 = 2.1528$ and perilune $r_p = 0.1446$ [nd], and are shown in Fig. 11.

The convergence basin results for the Earth-Moon L_2 Lyapunov family are given in Fig. 7. Tori branches around the smaller subset of L_2 Lyapunov orbits with larger perilunes reach higher values ε than the larger subset, though this is not for all perilunes as it is for the L_1 Lyapunov orbits. Same as the L_1 Lyapunov family, all tori family branches examined show a sharp decrease in convergence basin near the frequency resonance $\omega_1/\omega_0 = 1/4$, this time corresponding to roughly $r_p = 0.0230$ [nd]. Again, no failures modes of collapse were seen near this drop. The constant ω_1 branch of tori exists around nearly all periodic orbits in the analysis. The stabil-

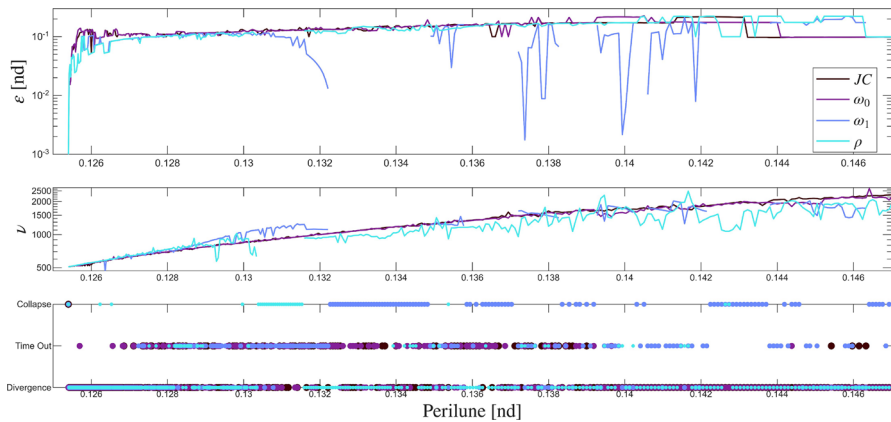


Fig. 6 Earth-Moon CR3BP L_1 vertical invariant tori convergence basins

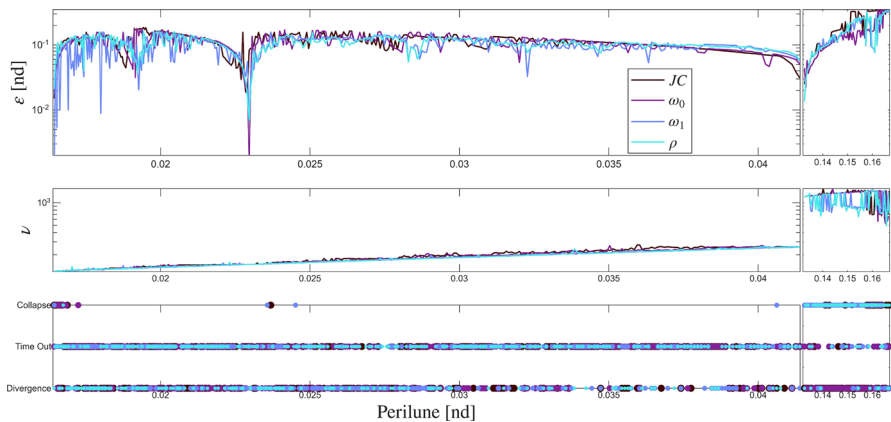


Fig. 7 Earth-Moon CR3BP L_2 Lyapunov invariant tori convergence basins

ity indices are notably different in the smaller subset, when compared to the larger. Just as was the case for tori around the smaller subset of L_1 Lyapunov orbit families, the constant ρ failure modes are dominated by collapse. Failure modes for the larger subset are more evenly distributed. The only region of distinct and consistent convergence basin difference occurs in the larger subset between perilunes of 0.0379 and 0.0414, where the constant ω_1 and ρ branches hold an average basin measure of $\Delta\varepsilon = 0.009$ [nd] more than the constant ω_0 and JC branches. This is the same qualitative location as was found in the L_1 Lyapunov family analysis. Tori family branches were continued out of a single L_2 Lyapunov orbit with frequency $\omega_0 = 1.8594$ and perilune $r_p = 0.1553$ [nd], and are shown in Fig. 11.

The convergence basin results for the Earth-Moon L_2 halo family are given in Fig. 8. This family does not show a consistently distinct region where one tori family branch's convergence basin exceeds the rest, or where one manifold branch exceeds the other within the stable band. Similar trends to the L_1 halo family's convergence

basins exist: the constant ω_1 branch does not exist in CM-B, the convergence basins of CM-A and CM-B both decrease as they approach the stable band from either side, and CM-A has overall larger convergence basins because it is farther from a singularity. The L_2 halo family does have more drop offs in basin measure with two noticeable in CM-B at $\{r_p = 0.0210, \omega_1/\omega_0 = 4/5\}$ and $\{r_p = 0.0309, \omega_1/\omega_0 = 3/4\}$, and one in CM-A at $\{r_p = 0.0659, \omega_1/\omega_0 = 1/2\}$. Near the stable band in CM-A there is a distinct stability index difference at the convergence basin edge where again the constant ω_1 branch does not exist. The constant JC branches are more unstable while the constant ω_0 branches are more stable. The constant ρ branch mimics the stability indices of the underlying periodic orbit. Near the bifurcation with the L_2 Lyapunov family, the constant JC tori branches fail in collapse. This cluster of collapse failure mode was present in the same qualitative location as the L_1 halo family, however, with the constant ρ branch. The constant ρ branch for the L_2 halo family is dominated by divergence failure. Tori family branches were continued out of a single L_2 halo orbit with frequency $\omega_0 = 2.0654$ and perilune $r_p = 0.0862$ [nd], and are shown in Fig. 11.

Lastly, the convergence basin results for the Earth-Moon L_2 vertical family are given in Fig. 9. The family shows a clear distinction in convergence basin between the constant JC branch, and all other branches between perilunes 0.1540 and 0.1668 corresponding to underlying orbital frequencies of 1.7367 and 1.7832, respectively. Another notable result is the dip in basin measure by all families near $r_p = 0.1441$ which corresponds to $\omega_1/\omega_0 = 1/10$. The stability indices of constant JC and ω_0 largely mimic the underlying periodic orbit stability, while constant ρ and ω_1 branches oscillate. Collapse of constant ρ branches occurs near L_2 , with other failure modes more evenly distributed throughout. Constant ω_1 branches exist almost everywhere in the L_2 Vertical family, in contrast to the L_1 family. This is likely because of the difference in potential well between the libration points coupled with the dependence on both free tori parameters of the ω_1 branch constraint. Tori family branches were continued out of a single L_2 vertical orbit with frequency $\omega_0 = 1.7654$ and perilune $r_p = 0.1611$ [nd], and are shown in Fig. 11.

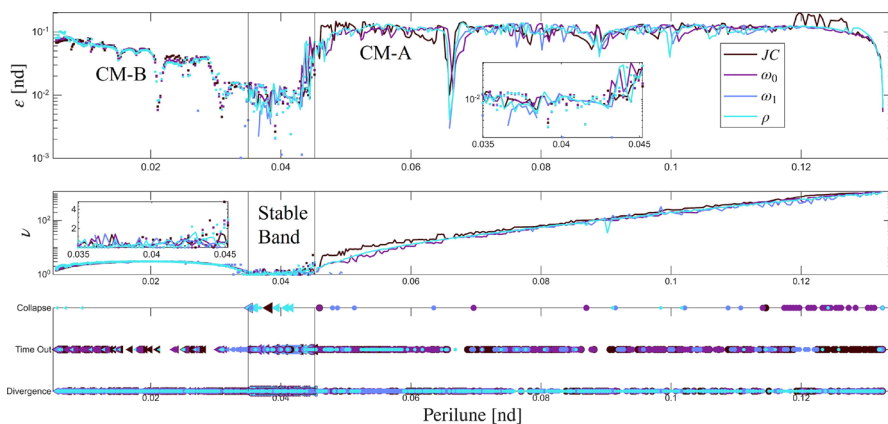


Fig. 8 Earth-Moon CR3BP L_2 halo invariant tori convergence basins

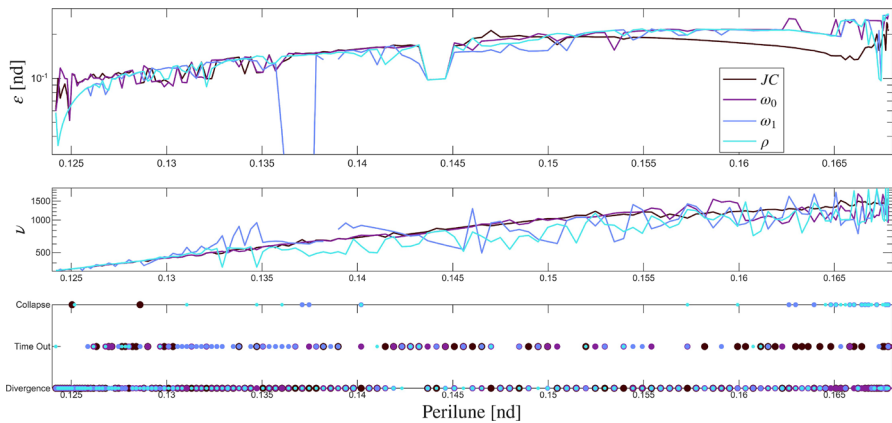


Fig. 9 Earth-Moon CR3BP L_2 vertical invariant tori convergence basins

Overall, tori branches generated from the smaller set of both Lyapunov orbit families showed the largest convergence basins because they are nearest to the local linear regions of the respective libration points and away from any singularities. Corresponding orbit families across the two libration points showed similar trends in overall values ε , but varied in where certain tori family branches were better approximated by the linear parametrization. Several basin sinks were identified near resonant tori frequencies; slope of the frequency space directly impacts the convergence basin. Constant ω_1 branches only exist in certain subsets of each orbit family, which in general do not necessarily correspond with global stability characteristics or proximity to resonance. Of the six orbit families analyzed, only the L_2 halo family neglected to show a region where the linear parametrization of 2-dimensional tori better approximated certain branches.

In order to capture the overall convergence basin behavior for each periodic orbit family, a polynomial fit for each basin edge was generated by averaging all family branch data points. Family branches that did not exist around particular periodic orbits were ignored. A quintic polynomial was used to capture sufficient trends in basin measure. The polynomial takes the following form with perilune distance as the independent variable.

$$\varepsilon(r_p) \approx \sum_{i=0}^5 c_i \cdot (r_p)^i \quad (37)$$

Separate polynomial fits were computed for each manifold branch of both halo families. Results are visualized in Fig. 10. Fits are not shown for the inner Lyapunov subfamilies because of the large gap in perilune where QPITs do not exist. As evident from Figs. 4 and 7, these regions, once averaged over all family branches, can be well approximated by straight lines.

Invariant curve families of each tori family branch are grown from an orbit from each family in Fig. 1 who exhibited noticeable differences in convergence basin mea-

sure to expose further geometric differences. They appear in Fig. 11 projected into configuration space with the linear approximation corresponding to $\varepsilon = 0.02$ illustrated as a grey disk. Note that the linear approximation for the Lyapunov families is a zero width ellipse in the z direction. The fixed point of the periodic orbit is plotted as a black circle.

From a quick glance, it is easy to identify the difference in geometry between the L_1 and L_2 Lyapunov invariant curve families. This is because the underlying periodic orbits belong to different subsets of the periodic families that illicit 2-dimensional QPITs in their vicinity, with branches from the L_2 Lyapunov orbit being closer to the libration point. Within the L_2 Lyapunov tori branches around this specific PO, the constant JC had the largest convergence basin. From the invariant curve geometries this tori branch is clearly different from the other three as it continues into the $-x$ direction, rather than $+x$. The L_1 halo orbit selected also had its constant JC branch show the largest convergence basin. Likewise, this branch's invariant curve geometry is starkly different, continuing by contrast in the $-z$ direction. The L_1 halo constant ω_0 branch eventually folds into itself and becomes multi-lobal. Interestingly, this also happens around the L_2 halo orbit but with the constant JC branch. For tori grown from both halo orbits, the constant ρ family branch can be continued the farthest away from the underlying periodic orbit. The constant ω_1 and ρ branches are much smaller around the L_1 vertical orbit. These same family branches display much larger convergence basins. Finally, the constant JC branch from the L_2 vertical orbit grows noticeably less in the x and z directions than the others, while its convergence basin is also distinctly smaller.

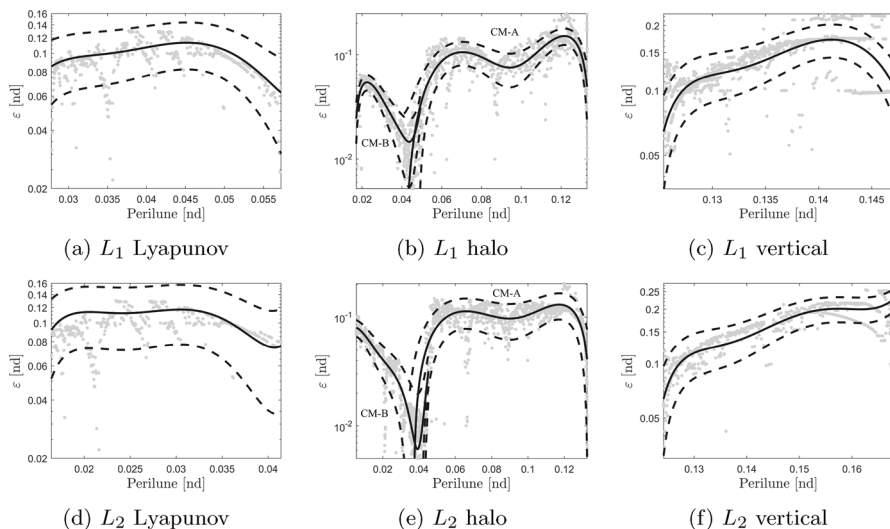


Fig. 10 Quintic polynomial fits of 2-dimensional tori convergence basin measure over averaged family branch data points for periodic orbit families in the Earth-Moon CR3BP. 95% confidence intervals are plotted with dashed lines. Inner Lyapunov subfamily data is not shown

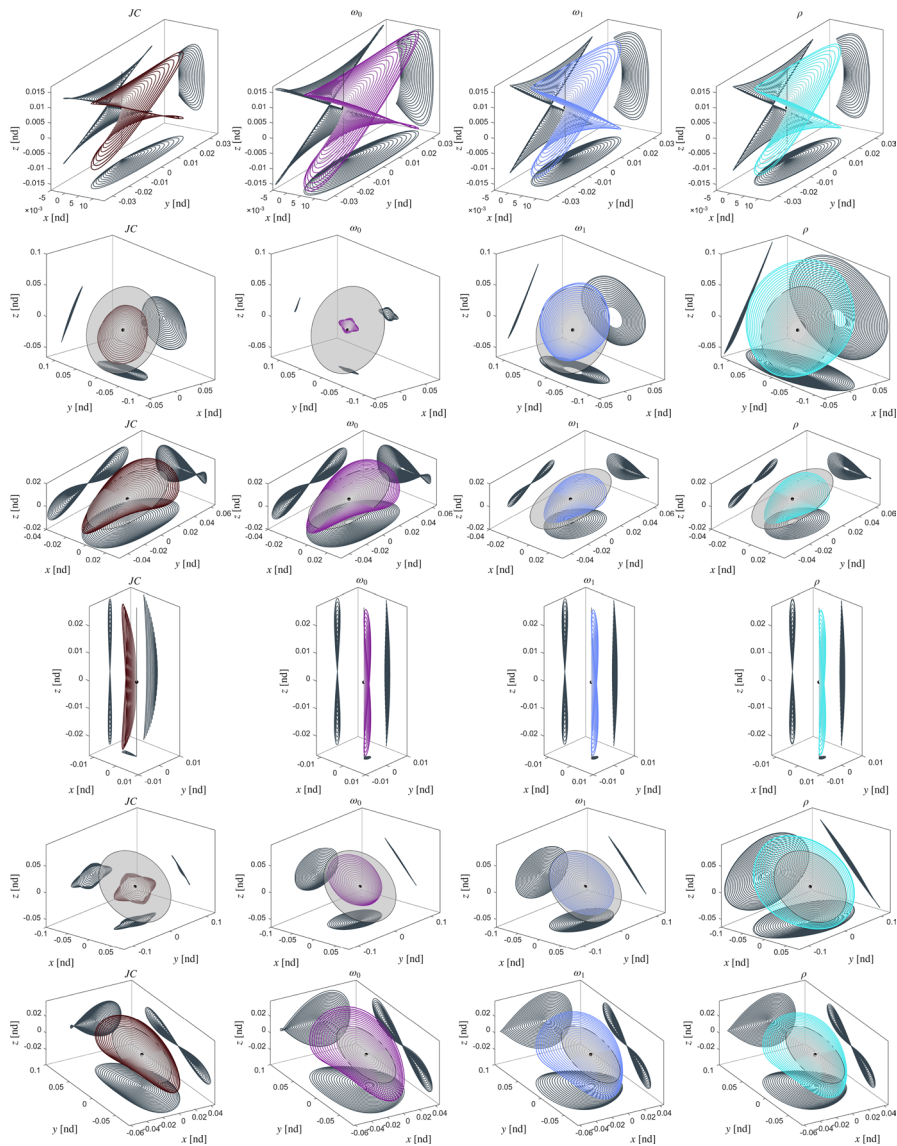


Fig. 11 1-Dimensional invariant curve families around select periodic orbits in the Earth-Moon CR3BP. From top to bottom row: L_1 Lyapunov, L_1 halo, L_1 vertical, L_2 Lyapunov, L_2 halo, and L_2 vertical. Each linear approximation appears as a grey disk with $\varepsilon = 0.02$. For corresponding frequency and perilune data, see Sec. 5.1

5.2 Variation of μ in the CR3BP

The effects of varying the CR3BP's singular parameter μ on the convergence basin size are now examined for tori generated around both the L_1 and L_2 libration points. The frequency and configuration space characteristics, as well as the convergence

basin measures of the L_1 and L_2 3:1 halo orbits are shown in Fig. 12 as μ is varied. Both orbits work back towards the larger primary body while growing in size as μ increases. With ω_0 constant for these gravitational parameter families, ω_1 increases for L_1 compared to a decrease for L_2 .

The convergence basin measures of all tori branches that exist for both μ orbit families grow as μ increases, though the L_2 family begins this trend near $\mu = 0.009$. This is primarily due to the overall growth of the underlying periodic as the mass parameter increases. Constant ω_1 branches exist for small subsets of the parameter for both cases. For μ values between 0.033 and 0.05, the constant ω_0 branch of tori for the L_1 3:1 halo orbit presents a consistently smaller basin measure ε , indicating that the linear parametrization is a poor representation of the structure when compared to the other branches. The convergence basins around the L_2 3:1 halo orbit are largely the same across different tori branches. This is consistent with the results for the entire Earth-Moon L_2 halo family depicted in Sec. 5.1. The basin measures do however dip near $\mu = 0.0036$. Unlike the sharp spikes seen in the Earth-Moon CR3BP data, this dip does not correlate with a low integer resonance.

5.3 Variation of e in the ER3BP

Lastly, the effects of varying the eccentricity parameter e on convergence basin size in the ER3BP are investigated. Specifically, constant ω_0 tori in the vicinity of all $q = 1$

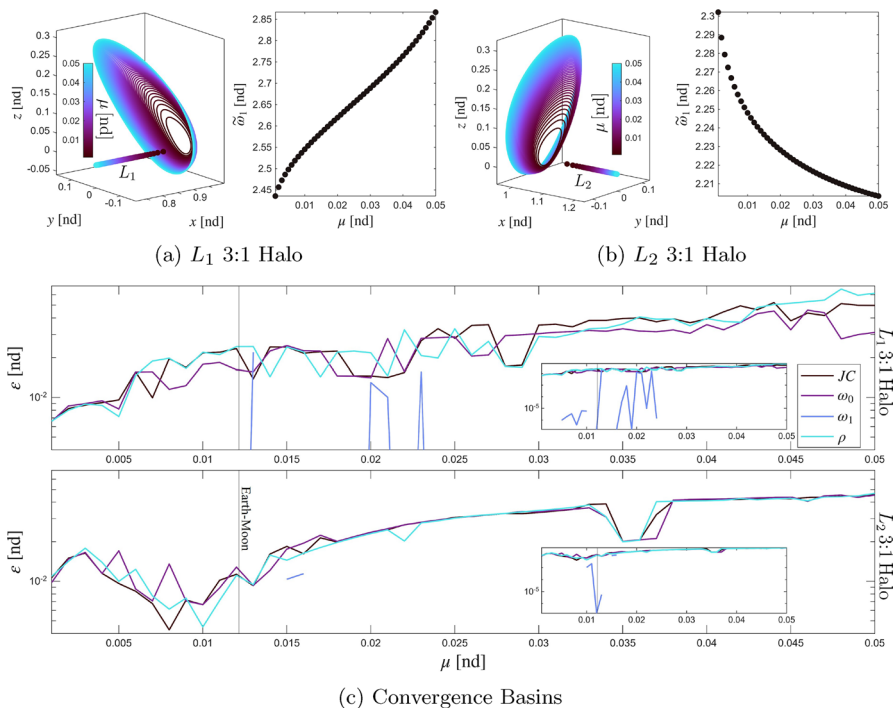


Fig. 12 Convergence basin data from tori grown out of 3:1 sidereal resonant halo orbits in the CR3BP for varying values μ

sidereal resonant L_2 halo orbits with $f_0 = 0$ at secondary apoapse are examined for their significance in mission design and notoriety in ephemeris transition. For this analysis, the mass parameter is held constant at the Earth-Moon value $\mu = 0.01215$ while e is varied. The resulting families of periodic orbits and their frequency characteristics are displayed in Fig. 13.

Because of the independent variable change between the CR3BP and ER3BP, the frequencies at $e = 0$ in Fig. 13 don't match their counterparts in Fig. 2, as the underlying period of each orbit is now increased to an integer multiple of the dynamic forcing frequency. Frequencies are labeled by their associated subspace manifold, with CM-A corresponding to the complex center component that exists in the CR3BP, and CM-B corresponding to the complex center component that bifurcates from the trivial pair of unitary multipliers as e moves away from zero. Both the 2:1 and 4:1 resonant orbits then have 3-dimensional QPITs in their vicinity for all values $e \in (0, 0.1]$. Though not shown, the frequency structure of the same orbits grown with $f_0 = \pi$ at apoapse is qualitatively the same. The 3:1 resonant orbit, which lies within the zone that sees difficulty transitioning to ephemeris [17, 44], has a more complicated frequency structure. A torus bifurcation occurs near $e = 0.0455$ where an additional frequency corresponding to CM-C enters the picture. This initiates a stable (elliptic) band in the eccentricity family of periodic orbits. Here, 4-dimensional QPITs exist and there are 3 independent sub-families of constant ω_0 branches of 2-dimensional tori that can be explored. Later, another torus bifurcation occurs near $e = 0.0468$ where CM-C collides with CM-A, leaving only CM-B and 2-dimensional QPITs

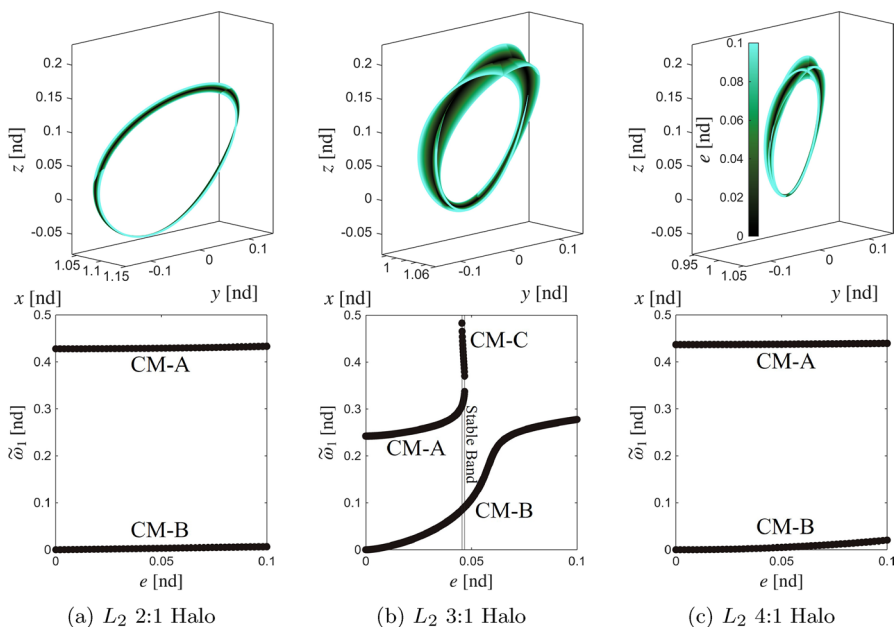


Fig. 13 Frequency data and configuration space geometry of $q = 1$, L_2 sidereal resonant halo orbits corresponding to $f_0 = 0$ at secondary apoapse in the ER3BP for varying values e and $\mu = 0.01215$. All frequencies are in [rad/rad].

behind. For the same 3:1 resonant orbit grown instead with $f_0 = \pi$ at apoapse (not shown), there is no stable band, and all higher dimensional tori are destroyed by $e = 0.0114$, where then the periodic orbit becomes purely hyperbolic.

The invariant curve geometry of tori in the ER3BP is generally more complex than similar structures in the CR3BP. To capture this, the number of discretized invariant curve points was increased to $N = 50$. Only constant w_0 branches of 2-dimensional tori can be computed because of the prescribed sidereal resonance criteria. The convergence basin results for the L_2 2:1 halo family are given in Fig. 14. For $e > 0.0448$, the basin measure of CM-A is consistently larger than CM-B. Despite relatively similar basin measures, the stability characteristics of the tori computed at basin edges are starkly different between the two manifolds. The stability variations for CM-B are much smaller than CM-A. This is because tori generated from CM-B originally bifurcate from the trivial pair of multipliers in the CR3BP, and in this eccentricity range, invariant curves grown in this manifold extend along the periodic orbit tangent itself. Little significant geometry change correlates with little significant stability change. Divergence is the dominant failure mode over the entire eccentricity range.

The convergence basin results for the L_2 3:1 halo family are given in Fig. 15. CM-B shows a larger convergence basin for $e \in [0.0231, 0.0290]$ and $e > 0.0418$. As the first torus bifurcation is encountered, the basin measures of CM-A and CM-C drop significantly. Overall, the average value ε for the 3:1 resonant orbit is an order of magnitude less than that of the 2:1 resonant orbit. This behavior is consistent with [17, 44]. Stability of all three manifolds follow the trend of the underlying periodic orbit, with most deviation seen in and near the stable band. Within the stable band, the only failure mode seen was divergence, and in general time out failures were rare.

The convergence basin results for the L_2 4:1 halo family are given in Fig. 16. Unlike the other two resonant orbit eccentricity families, the tori grown from the 4:1 see a significant steady decline in basin measure as eccentricity increases. This is a result of repeated close encounters with the singularity at the second primary; the underlying periodic orbit now makes four revolutions in one primary period. After overcoming the basin deficiency caused by the bifurcation from the resonant fre-

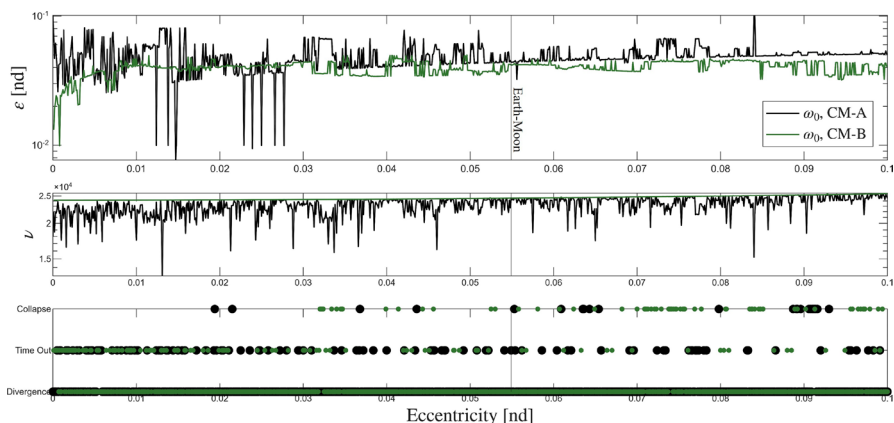


Fig. 14 L_2 2:1 halo invariant tori convergence basins in the ER3BP for $\mu = 0.01215$

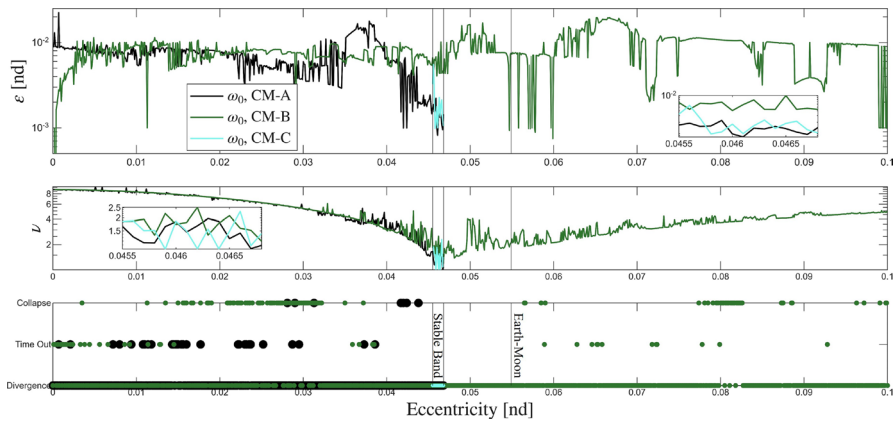


Fig. 15 L_2 3:1 halo invariant tori convergence basins in the ER3BP for $\mu = 0.01215$

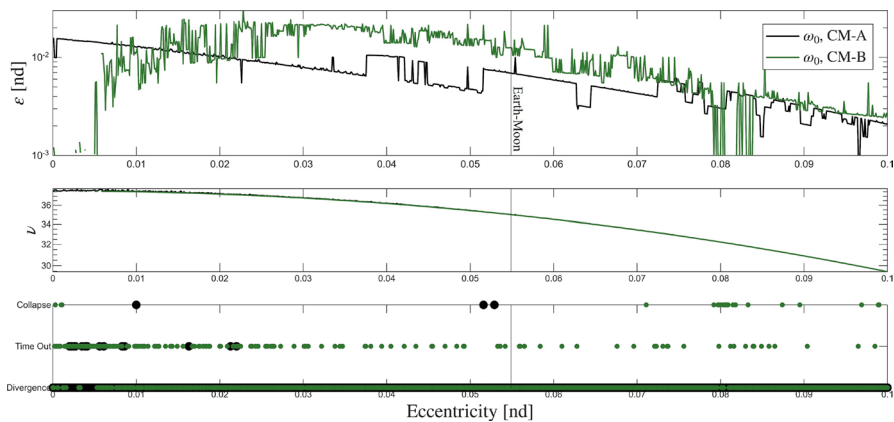


Fig. 16 L_2 4:1 halo invariant tori convergence basins in the ER3BP for $\mu = 0.01215$

quency 0:1 at $e = 0$, CM-B consistently shows a larger basin measure for $e > 0.0221$. Deviations in stability index from the underlying periodic orbit are minimal for both manifolds compared to the total deviation in stability index of the eccentricity family. Divergence again was the dominant failure mode, particularly for CM-A. CM-B was susceptible to time-out throughout the eccentricity range.

Overall, no trend was observed that had consistently smaller or larger basin measures for the manifold bifurcating out of the trivial pair of multipliers. Still, there were clear ranges in eccentricity that had dependable differences in basin measure between the manifolds. Compared to the Earth-Moon L_2 halo CR3BP family, the eccentricity families in the ER3BP for the select L_2 resonant halo orbits displayed far fewer failures in time out. This is attributed to the time-dependent nature of the ER3BP: when converging towards a solution for a particular tori, an element of freedom in the solution space of the structure is taken away because θ_0 is now phase locked to the vector field with an associated f_0 . As mentioned, the geometry of torus

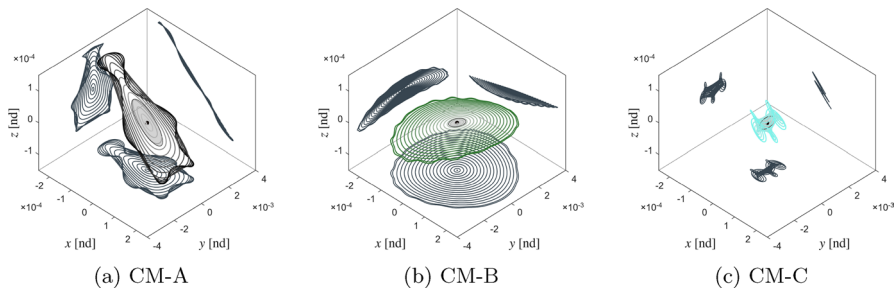


Fig. 17 Invariant curve families for each manifold of the L_2 3:1 sidereal resonant halo orbit in the ER3BP with $e = 0.0461$ and $\mu = 0.01215$. Each linear approximation appears as a grey disk with $\varepsilon = 1 \times 10^{-3}$

structures in the time-varying ER3BP is in general more complex than that found in the time invariant CR3BP. To illustrate these complexities, constant ω_0 family branches of 2-dimensional tori are grown until destruction from an L_2 3:1 halo orbit residing in the stable band with $e = 0.0461$ and $\mu = 0.01215$. The invariant curves of tori from each manifold are shown in Fig. 17, where the axes are not to scale to expose the intricacies. In particular, the invariant curves are those around the periodic orbit's apoapse fixed point corresponding to $f_0 = 0$. All tori branches shown can be continued two orders of magnitude less than similar structures in the CR3BP. This is likely because the stable band is riddled with fold bifurcations [44].

6 Conclusion

This work explored the connection between higher dimensional tori family branches and their linear parametrization through the use of computational convergence basins in both the CR3BP and ER3BP. In doing so, several trends were identified. Convergence basins around all orbit families in the CR3BP decrease as the orbits approach singularities in the dynamical system. Sharp drops in convergence basin measure were also seen when linear frequencies neared internal resonance. The constant ω_1 family branches of 2-dimensional tori failed to exist around many subsets of orbits examined. There are clear subsets of periodic orbit families where one tori family branch is consistently better or more poorly approximated by the linear parametrization, with the exception of the L_2 halo family in the Earth-Moon CR3BP. A similar statement can be made about eccentricity orbit families in the ER3BP, but with constant ω_0 tori along different manifolds. Lastly, the tori convergence basin measure decreased as a stable (elliptic) band was approached for orbit families that contained one. Polynomial fits of all convergence basin data collected in the CR3BP provide a practical reference for initializing these structures via the GMOS algorithm. This has the potential to reduce computation time in the numerical continuation of 2-dimensional tori families in mission design and analysis software.

Acknowledgements The Texas A&M High Performance Resource Computing Center is acknowledged for its role in data generation and processing. Beom Park is thanked for several discussions and insights into stability, and linear parametrizations.

Author Contributions Ian M. Down conducted all research and analysis, and wrote the full manuscript. Kathleen C. Howell, Manoranjan Majji, and Kyle T. Alfriend provided supervision, conceptual guidance, and input. All authors reviewed the manuscript.

Funding Partial funding for this collaboration was provided by the Hagler Institute of Advanced Study at Texas A&M University.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that there is no Conflict of interest associated with the work reported in this article.

References

1. Farquhar, R.W., Kamel, A.A.: Quasi-periodic orbits about the translunar libration point. *Celest. Mech.* **7**(4), 458–473 (1973)
2. Howell, K.C., Pernicka, H.J.: Numerical determination of lissajous trajectories in the restricted three-body problem. *Celest. Mech.* **41**(1–4), 107–124 (1987)
3. Gómez, G., Masdemont, J., Simó, C.: Quasihalo orbits associated with libration points. *J. Astronaut. Sci.* **46**, 135–176 (1998)
4. Gómez, G., Mondelo, J.M.: The dynamics around the collinear equilibrium points of the RTBP. *Phys. D* **157**(4), 283–321 (2001)
5. Jorba, A., Masdemont, J.: Dynamics in the center manifold of the collinear points of the restricted three body problem. *Phys. D* **132**(1–2), 189–213 (1999)
6. McCarthy, B., Howell, K.: Construction of heteroclinic connections between quasi-periodic orbits in the three-body problem. *J. Astronaut. Sci.* **70**(4), 24 (2023)
7. Bonasera, S., Bosanac, N.: Computing natural transitions between tori near resonances in the earth–moon system. *J. Guid. Control Dyn.* **46**(3), 443–454 (2023)
8. Barden, B.T., Howell, K.C.: Fundamental motions near collinear libration points and their transitions. *J. Astronaut. Sci.* **46**, 361–378 (1998)
9. Howell, K.C., Marchand, B.G.: Natural and non-natural spacecraft formations near the L1 and L2 libration points in the sun-earth/moon ephemeris system. *Dyn. Syst.* **20**(1), 149–173 (2005)
10. Duering, M., Vasile, M., Landgraf, M.: Uncontrolled spacecraft formations on two-dimensional invariant tori. In: 23rd International Symposium on Spaceflight Dynamics (2012)
11. Collange, G., Leitner, J.: Spacecraft formation design near the sun-earth L2 point. In: AIAA Guidance, Navigation, and Control Conference and Exhibit, p. 4781 (2004)
12. Down, I.M., Majji, M., Howell, K.C.: On-manifold low-thrust rephasing of quasi-periodic orbits. arXiv preprint (2025). <https://arxiv.org/pdf/2507.07940>
13. Down, I.M.: Manifold Constrained Trajectory Optimization of Quasi-Periodic Orbits. Ph.D. thesis, Texas A&M University (2025)
14. Lujan, D., Scheeres, D.J.: Earth–moon L2 quasi-halo orbit family: characteristics and manifold applications. *J. Guid. Control Dyn.* **45**(11), 2029–2045 (2022)
15. Lujan, D.A.: Methods to Explore Families of Quasi-Periodic Orbits with Applications in Astrodynamics. Ph.D. thesis, University of Colorado at Boulder (2023)
16. McCarthy, B.P., Howell, K.C.: Four-body cislunar quasi-periodic orbits and their application to ballistic lunar transfer design. *Adv. Sp. Res.* **71**(1), 556–584 (2023)

17. Park, B., Howell, K.C.: Assessment of dynamical models for transitioning from the circular restricted three-body problem to an ephemeris model with applications. *Celest. Mech. Dyn. Astron.* **136**(1), 6 (2024)
18. Kelly, P.M.: Quasi-Periodic Motion in the Earth–Moon–Sun Bi-Elliptic Restricted Four-body Problem. Ph.D. thesis, Texas A&M University (2024)
19. Grebow, D.: Generating Periodic Orbits in the Circular Restricted Three-body Problem with Applications to Lunar South Pole Coverage, pp. 27–93. MSA thesis, School of Aeronautics and Astronautics, Purdue University (2006)
20. Lujan, D., Scheeres, D.J.: Dynamics in the vicinity of the stable halo orbits. *J. Astronaut. Sci.* **70**(4), 20 (2023)
21. Ferrari, F., Lavagna, M.: Periodic motion around libration points in the elliptic restricted three-body problem. *Nonlinear Dyn.* **93**, 453–462 (2018)
22. Broucke, R.: Stability of periodic orbits in the elliptic, restricted three-body problem. *AIAA J.* **7**(6), 1003–1009 (1969)
23. Perko, L.: *Differential Equations and Dynamical Systems*. Springer, Heidelberg (2013)
24. Broer, H.W., Huitema, G.B., Sevryuk, M.B.: *Quasi-Periodic Motions in Families of Dynamical Systems: Order Amidst Chaos*. Springer, Heidelberg (1996)
25. Jorba, A., Villanueva, J.: On the persistence of lower dimensional invariant tori under quasi-periodic perturbations. *J. Nonlinear Sci.* **7**, 427–473 (1997)
26. McCarthy, B.: *Cislunar Trajectory Design Methodologies Incorporating Quasi-periodic Structures with Applications*. Ph.D. thesis, Purdue University (2022)
27. Kelley, W.G., Peterson, A.C.: *Difference Equations: An Introduction with Applications*. Academic Press, New York (2001)
28. Lujan, D., Scheeres, D.J.: Global L2 quasi–halo family and their characteristics. In: *Proceedings of the AAS/AIAA Space Flight Mechanics Conference* (2021)
29. Sansottera, M., Locatelli, U., Giorgilli, A.: A semi-analytic algorithm for constructing lower dimensional elliptic tori in planetary systems. *Celest. Mech. Dyn. Astron.* **111**(3), 337 (2011)
30. Schilder, F., Osinga, H.M., Vogt, W.: Continuation of quasi-periodic invariant tori. *SIAM J. Appl. Dyn. Syst.* **4**(3), 459–488 (2005)
31. Baresi, N., Olikara, Z.P., Scheeres, D.J.: Fully numerical methods for continuing families of quasi-periodic invariant tori in astrodynamics. *J. Astronaut. Sci.* **65**, 157–182 (2018)
32. Kolemen, E., Kaskin, N.J., Gurfil, P.: Quasi-periodic orbits of the restricted three-body problem made easy. In: *New Trends in Astrodynamics and Applications III* (AIP Conference Proceedings Volume 886), vol. 886, pp. 68–77 (2007)
33. Olikara, Z.P.: *Computation of Quasi-Periodic Tori and Heteroclinic Connections in Astrodynamics Using Collocation Techniques*. Ph.D. thesis, University of Colorado at Boulder (2016)
34. Fernández-Mora, Á., Haro, A., Mondelo, J.-M.: Flow map parameterization methods for invariant tori in quasi-periodic hamiltonian systems. *SIAM J. Appl. Dyn. Syst.* **23**(1), 127–166 (2024)
35. Kumar, B., Anderson, R.L., Llave, R.: Rapid and accurate methods for computing whiskered tori and their manifolds in periodically perturbed planar circular restricted 3-body problems. *Celest. Mech. Dyn. Astron.* **134**(1), 3 (2022)
36. Down, I.M., Howell, K.C., Alfriend, K.T., Majji, M.: Relative motion in the center subspace of periodic solutions to the CR3BP. In: *Proceedings of the AAS/AIAA Astrodynamics Specialist Conference* (2023)
37. Junkins, J.L., Singla, P.: How nonlinear is it? A tutorial on nonlinearity of orbit and attitude dynamics. *J. Astronaut. Sci.* **52**, 7–60 (2004)
38. Jensen, E.L., Scheeres, D.J.: Semianalytical measures of nonlinearity based on tensor eigenpairs. *J. Guid. Control Dyn.* **46**(4), 638–653 (2023)
39. York, C.E.: *Convergence Basin Analysis in Perturbed Trajectory Targeting Problems*. Ph.D. thesis, Purdue University (2023)
40. Sharan, S., Singla, P., Eapen, R.T., Melton, R.G.: Higher-order differential correction schemes for the two-point boundary value problem. *J. Guid. Control Dyn.* **48**(7), 1477–1491 (2025)
41. Hildebrand, F.B.: *Introduction to Numerical Analysis*. Courier Corporation, North Chelmsford (1987)
42. Gardner, J.P., Mather, J.C., Clampin, M., Doyon, R., Greenhouse, M.A., Hammel, H.B., Hutchings, J.B., Jakobsen, P., Lilly, S.J., Long, K.S., et al.: The James Webb space telescope. *Sp. Sci. Rev.* **123**, 485–606 (2006)

43. Gardner, T., Cheetham, B., Forsman, A., Meek, C., Kayser, E., Parker, J., Thompson, M., Latchu, T., Rogers, R., Bryant, B., et al.: Capstone: a cubesat pathfinder for the lunar gateway ecosystem (2021)
44. Park, B., Howell, K.: Characterizing transition-challenging regions leveraging the elliptic restricted three-body problem: L2 halo orbits. In: AIAA SCITECH 2024 Forum, p. 1455 (2024)
45. Boudad, K., Howell, K.C., Davis, D.C.: Analogs for earth–moon halo orbits and their evolving characteristics in higher-fidelity force models. In: AIAA SCITECH 2022 Forum, p. 1276 (2022)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.